

# CM500288 Principles of Programming

## Graded Assignment 2: Java Connect4

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### 1 Introduction

The assessment of this unit consists of two courseworks. This document provides the specification for the second coursework: **Connect4 in Java**.

You can use any Integrated Development Environment (IDE) for the development of your code, but your code must run on Java 17 (the default version used on repl.it) without requiring the installation of additional libraries, modules or other programs. Standard packages included with Java (e.g. `java.util`) are fine.

Questions regarding the coursework can be asked on Engage in the General Q&A Forum, or by messaging the unit convenors.

### 2 Learning Objectives

At the end of this coursework you will be able to

- Plan, organise and implement program code to support reuse and maintainability of a software project.
- Provide a critical review of a software artefact in terms of software quality, design, reuse and robustness, and offer solutions to correct issues encountered.

### 3 Connect4

For your first coursework, we asked you to investigate and replace a program that we supplied (SRPN). In this coursework you will need to analyse, design and extend a second program which you can find on Engage (Connect4).

The challenges in this coursework arise from the fact that the Java code supplied should allow a human to play the game Connect4 against a computer. Connect4 is described at [https://en.wikipedia.org/wiki/Connect\\_Four](https://en.wikipedia.org/wiki/Connect_Four). Please

ignore the “Rule variations” listed on Wikipedia. You should only implement the standard version of the game.

Connect4 is a game that can be played between a human and a computer player, on a 6x7 grid. The goal is to connect 4 tokens either diagonally, horizontally or vertically.

You have been given a program that tries to implement this game. Unfortunately, our given code:

- Doesn't compile or run.
- Is poorly designed (e.g. it is largely made up of a huge main function without comments). With everything we've learned about Object Oriented (OO) programming, we will need to redesign the code to adhere to these principles.

Your coursework is to reflect on this code and design a working solution that follows OO principles. To do this, you will need to satisfy Requirements 1 and 2 below.

## 4 Requirements

1. Write a report (approximately 2 A4 pages of single-spaced, 10 point font) describing the ways in which our code could be restructured to better reflect the fundamental OO concepts of modularisation and encapsulation. You must include a discussion of the extent to which you think that inheritance, abstract classes and interfaces might be used to improve our code. You may argue either for or against their inclusion.

In this report you should, briefly and in your own words, explain what each of these object oriented principles are. Give an example of each, using the Connect4 game, indicating what you'd need to change to adhere to these principles. You may choose to include diagrams to accompany your text descriptions.

2. Write a program which matches the functionality of the Connect4 game. Your code must allow 1 human player to play a complete game against 1 computer player. Your code must be commented appropriately.

To get started, you may wish (but are not required) to use our code as a starting point. If you do, you may want to start by commenting out large parts of our code and altering our `printBoard()` method. This approach has the advantage of giving you a relatively manageable starting point in the debugging process. You may also wish to tackle our `placeCounter()` method next. However, you can also choose not to use our code, and start a new program.

Your code should follow the object oriented principles you described in Requirement 1. You must not rely on external libraries, please complete this assignment with the use of standard libraries only.

Please submit your uncompiled code. This means the .java files only – we do not want your compiled .class files. If you are using an IDE (not replit), please do not submit the entire directory structure of your project: **you should only submit .java files in one folder.**

## 5 Submission Instructions

You must submit **two** files for this coursework:

1. A PDF file titled `Connect4-report.pdf` containing your report for Requirement 1.
2. A .zip file titled `Connect4-code.zip` containing the .java source files required to compile and run Connect4.

Your files must be submitted to the Engage assignment page by the submission deadline shown. Submissions received after this deadline will be capped at 40% if received within 5 working days. Any submissions received after 5 working days will be marked at 0%. If you have a valid reason for an extension, you must submit an extension request by following the appropriate process outlined on this page <https://engage.bath.ac.uk/learn/course/view.php?id=36&section=5> – unit leaders cannot grant extensions.

You must leave yourself time to download your file from Engage, extract it, and check that you have attached the correct file, with the content that you want to be marked. **You** are responsible for checking that you are submitting the correct material to the correct assignment.

**Please check Engage for the submission deadline.**

## 6 Assessment

### 6.1 Conditions

The coursework will be conducted individually. Attention is drawn to the University rules on plagiarism. While software reuse (with correct referencing of the source) is permitted, we will only be able to assess your contribution.

You may be called on to explain or demonstrate your program. Cases leading to such an explanation and/or demonstration include but are not limited to suspected plagiarism.

### 6.2 Marking Scheme

Below is a breakdown of marks, with descriptions on how each criteria is met.

| Criteria                                                    | Max Score | Description                                                                                                                                                                                                              |
|-------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Report – Max 30                                             |           |                                                                                                                                                                                                                          |
| Report on OO concepts in relation to restructuring our code | max 30    | The report should include a description of each OO concept, with examples of how to implement each, referring to the Connect4 code and an explanation of how the code could be restructured to adhere to these concepts. |
| Connect4 code – Max 70                                      |           |                                                                                                                                                                                                                          |
| Functionality – revised (working) Connect4 code             | 30        | Revised (working) Connect4 code that allows 1 human player to play against 1 computer player                                                                                                                             |
| Code quality                                                | max 30    | Code is written cleanly and simply with appropriate use of object-oriented programming principles, such as data encapsulation and generalisation, classes and robustness.                                                |
| Commenting and naming                                       | max 10    | Code is consistently commented throughout, appropriate variable and method names.                                                                                                                                        |

**Total: Max 100 marks.**

### 6.3 Grades

Marks and feedback will be made available within 3 working weeks of the submission.

The following is a guideline for marks:

- **Above 40%** – Pass.
- **50-69%** – Very good. (Pass/Merit range)
- **70%+** – Outstanding. (Distinction range)